

Ghosts and shadows- evidential contrafactuals in Georgian.

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1 Introduction

Propositional attitude predicates such as *believe, think, know, doubt, etc* express relations between attitude holders (individuals) and propositions (Hintikka, 1969).

Of these, certain predicates such as *know, realize, regret* have been argued to trigger a *factivity* inference, i.e they imply the truth of their complements.

- (1) a. Maia knows that it is raining
 $\rightsquigarrow$ It is raining.
- b. Maia believes that it is raining
 $\nrightarrow$ It is raining.

The factivity inference in predicates such as *know* are typically analysed as presuppositions i.e they encode backgrounded information, and they also project across various environments. They are often contrasted with veridical predicates which entail, but do not presuppose the truth of their complements.

- (2) a. Maia is right that it is raining.
 $\rightsquigarrow$ It is raining.
- b. Maia isn't right that it is raining.
 $\nrightarrow$ It is raining.
- (3) a. Maia knows that it is raining
 $\rightsquigarrow$ It is raining.
- b. Maia doesn't know that it is raining,
 $\rightsquigarrow$ It is raining.

2 What is (contra)factivity (Holton, 2017; Roberts and Özyıldız, 2025)

It has been suggested that natural language does not have predicates that presuppose the falsity rather than the truth of their complements, i.e. contrafactivities.

- (4) a. Maia contras that it is raining.
 $\rightsquigarrow$ It is not raining.
- b. Maia doesn't contras that it is raining,
 $\rightsquigarrow$ It is not raining.

(Roberts and Özyıldız, 2025)

Note, languages clearly have the means to express false belief reports using more complex phrases or additional affixes (eg- disprove, misperceive), but this claim questions the lack of contrafactivity being lexicalized as a single predicate such as *know*.

(Holton, 2017) outlines the following characteristics of a true contrafactive predicate:

1. Stative- Homogenous event structure
2. Monomorphemic- It must be lexicalized into a single predicate
3. Doxastic- The predicate must express a belief relation between the attitude holder and the complement
4. contrafactive inference, i.e presupposes the falsity of its complement

2.1 Stative

Excludes predicates like *wijsmaken* in Dutch/ *weismachen* in German. These predicates do trigger falsity inferences, similar to *delude/ deceive*.

- (5) a. Ze maakte me wijs dat ze rijk was...#en ze was wel rijk
she made me wise that she rich was...#and she was PRT rich.
She fooled me into thinking that she was rich... and she *was* rich.
- b. Ze maakte me niet wijsdat ze rijk was... ik wist al dat ze arm was.
She made me not wise that she rich was... I knew already that she poor was.
She didn't fool me into thinking that she was rich. I already knew that she was poor.

However, the falsity inference does not project under negation. Moreover, these predicates involve a change of state where the attitude holder undergoes a change from not believing p to believing p where p is the proposition denoted by the complement clause.

2.2 Monomorphemic

Excludes English predicates like *misremember* and *disprove* where the falsity inference is clearly a result of the negative prefix, or other adverbial modifiers such as *wrongly* or *mistakenly*.

- (6) a. The professor remembered that her cat could understand Latin.
 $\rightsquigarrow$ Her cat could understand Latin
- b. The professor misremembered/ remembered incorrectly that her cat could understand Latin.
 $\rightsquigarrow$ Her cat could not understand Latin.

2.3 Doxastic

Excludes predicates like *lie*, *pretend* where the attitude holder does not believe the truth of the complement clause. In these cases the falsity of the complement is entailed rather than presupposed. These predicates can embed false propositions with and without negation, but not true propositions.

- (7) a. Maia pretended that her cat could (# not) understand Latin.
 b. Maia didn't pretend that her cat could (# not) understand Latin.

These predicates do not ascribe a belief to the attitude holder but rather entail the falsity of the proposition.

- (8) Maia pretended that her cat could understand Latin.
 a. No she did not.
 b. # No it doesn't.

2.4 Contrafactive

There are two potential candidates of *true* contrafactives in the existing literature; eg. *se creerse* (Spanish), *yǐwéi* (Mandarin)

The falsity inference of *se creerse* cannot be cancelled by discourse, making it a potential contrafactive.

- (9) a. #I don't know whether it is raining or not, but John knows that it is raining.
 b. #No sé si esta lloviendo o no, pero Juan **se cree** que esta lloviendo.
 'I don't know if it's raining or not, but Juan REFL believes that it is raining.' (Anvari et al., 2019)

It has been observed that *yǐwéi* also triggers false belief inferences.

- (10) *Context: The speaker is pretending to be sick to skip school.*
 a. Māma yǐwéi wǒ bǐng le.
 Mother yǐwéi 1.SG sick ASP.
 Mother is under the impression that I am sick. (Glass, 2023)

However, the falsity inference in the case of *yǐwéi* has been argued to be defeasible, making it a predicate more along the lines of the English *believe*

- (11) rénmen yǐwéi tā shì yìwànfùwēng... ér tā dìquè shì
person.PL yǐwéi 3.s be billionaire... and 3.s indeed be.
People are under the impression that she is a billionaire.. and she actually is. (Roberts and Özyıldız, 2025)

se cree remains a potential candidate for a true contrafactive; it is a stative, doxastic predicate; and the falsity inference projects under questions, as well as the scope of a universal quantifier. However, mood alternations can affect the falsity inference.

- (12) a. Hilda se cree que Jirafales es honesto.
Hilda REFL believes that Jirafales is honest.
↪ Jirafales is not honest and Hilda believes that he is (Indicative)
- b. Hilda no se cree que Jirafales es honesto.
Hilda no REFL believes that Jirafales is honest.
↪ Jirafales is not honest and Hilda doesn't believe that he is (Indicative)
- c. Hilda no se cree que Jirafales sea honesto.
Hilda no REFL believes that Jirafales is.SUBJ honest.
↪ Hilda doesn't believe that Jirafales is honest. (Subjunctive) (Maldonado and Percus, 2025)

3 The Georgian contrafactive *moečvena*

stative

✓

doxastic

✓

mono-morphemic

×

falsity presupposition

✓

Stativity *moecvena* is incompatible with durative adverbials.

- (13) nino-s *(ori c'ut-i)/or c'ut-ši mo-e-čven-a rom c'vim-d-a
nino-DAT (two minutes/two minutes-in) PVB-PV-evidence-AOR.3.SG COMP rain-IMPf-AOR.3.SG
Nino [falsely] believed that it was raining *(for two minutes/ in two minutes).

Falsity inferences

(14) *Context: Nino heard the sound of sprinklers going off outside his house, so he thought it was raining.*

- a. nino-s mo-e-čven-a rom c'vim-d-a
nino-DAT PVB-PV-evidence-AOR.3.SG COMP rain-IMPF-AOR.3.SG
Nino [falsely] believed that it was raining. (~ *it was not raining.*)

It also has a stronger falsity inference than the belief predicate *egona*, which can also give rise to similar interpretations.

- (15) nino-s e-gon-a rom c'vim-d-a
nino-DAT PV-believe-AOR.3.SG COMP rain-IMPF-AOR.3.SG
Nino believed that was raining.

Falsity inference projects from under questions, as well as the scope of universal quantifiers.

- (16) a. nino-s mo-e-čven-a rom c'vim-d-a [tu ara]?
nino-DAT PVB-PV-evidence-AOR.3.SG COMP rain-IMPF-AOR.3.SG [POL]?
Did Nino [falsely] believe that it was raining? (~ *it was not raining.*)
- b. q'vela st'udent'-s mo-e-čven-a rom c'vim-d-a
Every student-DAT PVB-PV-evidence-AOR.3.SG COMP rain-IMPF-AOR.3.SG
Every student [falsely] believed that it was raining. (~ *Every student believed it was raining.*)

But does not project from under negation.

- (17) nino-s ar mo-e-čven-a rom c'vim-d-a. martla c'vim-d-a.
nino-DAT NEG PVB-PV-evidence-AOR.3.SG COMP rain-IMPF-AOR.3.SG. really rain-IMPF-AOR.3.SG
Nino did not [falsely] believe that it was raining. It was really raining.

Morphological complexity Georgian verbs are always morphologically complex. A verbal root typically combines with different preverbs, pre-radical vowels, as well as other aspectual markers to give rise to a wide range of meanings.

Unlike the prefixes *mis-/dis-* the preverb *mo-* does not encode negation.

1	2	3	4	5	6	7	8	9	10	11
pvb	agr	pv	root	voice	ts	caus	tam	tam	agr	agr
mo	e		cven							a
	3.sg		evidence					aor	3.sg	

Other Georgian predicates with *mo-*

- (18) maia-m mo-i-gon-a rom davaleba rtul-i iuo
Maya-ERG PVB-PV-opinion-AOR.3SG COMP homework.NOM difficult-NOM be.AOR.3SG
Maya recalled that the homework was difficult
- (19) masc'avlebel-i mo-i-txov-s rom bavšv-eb-i ga-čum-d-nen.
teacher.NOM PVB-PV-request-FUT.3.SG COMP student-PL-NOM PVB-TS-PRS.3PL
The teacher demanded that the children be quiet

4 The evidential flavour of *moëčvena*

The predicate *moëčvena* can take both nominal as well as clausal arguments.

- (20) a. nino-s k'at'a mo-e-čven-a, magram kva iq'o
Nino-DAT cat.NOM PVB-PV-evidence-AOR.3.SG, but stone be.AOR.3SG
Nino[falsely] believed she saw a cat, but it was a stone.
- b. nino-s mo-e-čven-a rom c'vim-d-a
nino-DAT PVB-PV-evidence-AOR.3.SG COMP rain-IMPf-AOR.3.SG
Nino [falsely] believed that it was raining.

However, *moëčvena* differs from other potential contrafactuals in that it can only be used to report events directly perceived by the attitude holder.

- (21) a. #nino-s mo-e-čven-a rom šota ocdaati c'lisa.
Nino-DAT PVB-PV-evidence-AOR.3.SG COMP Shota thirty years.BE.PRS.3.SG.
Nino [falsely] believed that Shota is thirty years old.
- b. #nino-s mo-e-**gon**-a rom šota ocdaati c'lisa.
Nino-DAT PVB-PV-believe-AOR.3.SG COMP Shota thirty years.BE.PRS.3.SG.
Nino [falsely] believed that Shota is thirty years old.

Maldonado and Percus (2025) decompose *se creese* into the belief predicate *creer* and the operator **WRONGLY** in (22). This operator combines with the belief predicate to return a function whose domain only consists of worlds where the belief is false. They attribute the presence of wrongly to the reflexive *se*.

I propose a similar analysis for the preverb *mo-*

$$(22) \quad \llbracket mo- \rrbracket(P) = \lambda x. \lambda w : P(x)(w) \text{ is defined and } B_p(w) = 0 \text{ (Adopted from Maldonado and Percus (2025))}$$

Given the direct evidentiality restriction, I adopt the following denotation in (23) for the verbal root.

$$(23) \quad \llbracket cven(f)(B)(w)(\phi) \rrbracket = \text{is only defined if for all worlds } w', w' \in B(w) \\ \text{iff } f \text{ the perceived evidence in } w \text{ holds in } w' \text{ and } f \text{ is a choice function of type } \langle st, st \rangle \\ \text{such that } f(B(w)) \subseteq B(w).$$

If defined, $\llbracket cven(f)(B)(w)(\phi) \rrbracket = 1 \text{ iff for } \forall w' \in f(B(w)) : \llbracket \phi(w') \rrbracket = 1$

(Adopting the denotation for a direct evidential from Matthewson et al. (2006) for St'at'imcets)

5 Conclusion

Is *moečvena* a true contrafactive?

It is a stative, doxastic, morphologically complex predicate that gives rise to false belief inferences.

The falsity inference does seem to project from questions, as well as the scope of quantifiers. However, it *does not project from negation*, a major argument against it being a true contrafactive presupposition.

However, it is still the case that *moečvena* does give rise to stronger false belief inferences compared to more closely related predicates such as *e-gon-a* (to believe) and *c'armoisaxa* (to imagine) that are easily explained by analyzing them as contrafactives.

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